Sliding-Scale Autonomy for Shared Control Robotics in a Pick-and-Place Task

Chris Robinson*, Student Member, IEEE, Indika B. Wijayasinghe*, Member, IEEE, Dan O. Popa*[1], Senior Member, IEEE

***Abstract*—As robotic agents become increasingly present in human environments, task completion rates during human-robot interaction is an important topic of research. Safe collaborative robots executing tasks under human supervision often augment their perception and planning capabilities through traded control schemes. In this paper, we introduce a sliding scale model of teleoperative intervention, in which the user is allowed to select the proportion of the task space which is manual versus automatic. The proposed sliding scale replaces the traditional fixed mode in which sub-tasks are assigned to either a human operator or the robot. The advantage of the sliding scale is that human operators can better adapt to the complexity of real world environments, and make use of the autonomous capabilities of the robot while maintaining a high task completion rate. We experimentally tested this model of control against a traditional traded control scheme in a pick-and-place task. Results show that users who are allowed to adjust the intervention tier perform at the same success rate as the best-in-class fixed-mode comparison. Moreover, they experience task completion time improvements over an average performance with fixed-mode intervention.**

I. Introduction

As autonomous manipulators become increasingly more sophisticated, applications for use in environments beyond the traditional industrial material-handling niche have opened up. Of particular interest to this investigation are automated assistant-type robots which are capable of operating in human environments while performing tasks which are difficult or unpleasant for the operator to accomplish.

Robotic systems that share immediate spaces with humans in living environments pose a unique set of safety challenges because of the complexity of unstructured environments [1], but recent advances in collaborative robotics and advanced perception algorithms has driven the adoption of robotic systems into new venues. This expansion presents a prime opportunity to alleviate difficulties for humans on jobs that can be augmented by robotic tools. Autonomous systems can reduce cognitive loading from repetitive or focus-demanding tasks [2]. Furthermore, there are many cases, such as hazardous materials handling, for which it is impractical or unwise for humans to perform the task in question[3].

However, despite the benefits robots can offer, there are still significant gaps in their capabilities. Many researchers have proposed to supplement robot autonomy with human perception and decision making, effecting a traded-control scheme [4]. Given that there are advantages to certain parts of each participants repertoire, is it natural to divide the task load to humans and robots in accordance with their optimal capabilities. On the other hand, when the operator and robot both contribute to a teleoperation task achievement concurrently, the operational paradigm is defined as shared control [5]. The designation of which subtasks, and what proportion of total work should be assigned to human or robot refers to the level of autonomy of the system [6], a topic that has been increasingly studied with the advent of autonomous vehicles, the DARPA Robotics Challenge, and so on.

Though the full topic of shared control teleoperation carries an implicit dependence on partitioning task spaces into human and machine components, in this paper we focus on the specific domain of adjustable autonomy, meaning contextually discussions of systems for which the degree of control the operator asserts is not assumed to be fixed throughout the task.

In [7] and [8], research pertaining to human intervention is presented with the example case of bin-picking which resolved errors by elevating control to a human operator in the case that a fault is detected. These examples employ only after-the-fact error detection, a single autonomy level, and primarily focus on intervention by way of fault analysis.

The work in [2] presents a study on the quality of interaction between human users and an autonomous vehicle related to the level of autonomous action present in the vehicle. The paper presents user studies to assess the impact of different levels of autonomy to the quality of autonomous driving.

An explicit discussion of the impact of autonomy level in the context of human-robot cooperative tasks is given in [9], in which the authors describe a variable autonomy system based on the concept of action and neglect. In the absence of teleoperated input from the operator, an autonomous system function was triggered. The autonomy switching trigger was explored in the context of delay between human input and robot, rather than environmental observation.

Additionally, work in [10] presents a novel input device which interfaces a human to a robot using a novel input device which incorporates variable autonomy into its control structure. This paper focuses on aspects of the device as an input system, and in that context approaches the topic of differing levels of autonomy, but without expressly assigning the user control over the autonomy level, rather

[1]The authors are with Next Generation Systems Group. Dept. of Electrical and Computer Engineering, University of Louisville, KY, 40292, USA., Email:christopher.robinson.2;indika.wijayasinghe; dan.popa@louisville.edu

working to address the issues associated with changing input characteristics due to shifts in autonomy level.

In complex, changing environments, the range of tasks is sufficiently broad that a general-purpose robot would need a multitude of bespoke modules to adequately function. Furthermore, the rapidly changing workspace conditions may often render a given task model irrelevant in a momentary context and give rise to certain situations in which the robots autonomous capabilities are more likely to degrade. It is also possible that over-reliance on operator interventions may unduly tax certain untrained users, introducing inefficiencies to the system. It is therefore appropriate to adjust the ratio of human-driven tasks to robot tasks in accordance with the user and the current situation [11].

There are other similar examples, in which autonomy is positively correlated with speed and negatively correlated with accuracy, as in the case of human-computer interface performance expressed by Fitts' law [12]. Fitts' law establishes that there is an inverse relationship between the complexity of a task and the speed, presuming a constant 'information throughput', IP. In our case, IP can be likened to task completion success, and so we expect a similar conditional relationship between task complexity and the corresponding success threshold.

We therefore hypothesize that optimal performance of the human-robot system occurs at an intermediate level of autonomy, and that users are quite capable of determining appropriate levels of autonomy..

In this paper we investigate this hypothesis in the context of teleoperation, and propose a novel human-robot Variable Autonomy Planner (VAP). VAP elevates the human operator to a meta-level of control authority via a novel sliding-scale model for task accomplishment. In this model, the operator selects from a range of autonomy levels spanning from full teleoperation control of each robot movement up to complete autonomy prior to performing a task, based on their judgement of the difficulty thereof.

Each task is decomposed into a hierarchy of basic actions, for which either an autonomous or manual control mode can be used. By changing the degree of autonomy, the operator can select the proportion of automatic action the robot will take on a case-by case basis.

We experimentally investigate the effectiveness of this concept by testing 12 subjects on a robotic pick and place task. Our experiments indicate that in most cases fixed-autonomy levels have worse over-all performance with regard to task completion time and failure rates than the case in which our VAP algorithm selects the autonomy level. By contrast, when operators are allowed to select the level of autonomy, they can make an educated assessment of the problem and select superior balances of speed and quality through module choice. The paper is organized as follows:

In Section II, we describe the problem class in detail, formulate our VAP algorithm, and the metrics by which we gauge performance. In Section III, we outline the implementation case and experiment procedure we use for testing, as well as detail the hardware and software used in these trials. In Section IV, we present the results of our experiments, and analysis of them; Section V concludes the paper with a summary and plans for future extensions of these experiments.

II. PROBLEM DESCRIPTION & VAP ALGORITHM

To solve the problem of context-dependence of optimal autonomy level, we propose a methodological algorithm design which implements variable autonomy by preparing manual and automatic modules for each subtask the robot must perform. We then implement an additional decision on module choice based on a newly defined autonomy level parameter (AL).

At the outset of design, we consider a teleoperation task T, that can be decomposed into a set of subtasks $\{T_1, T_2, T_3...\}$ Furthermore, to execute the task, the robot must possess some elevated planning structure which handles transitions between subtasks. We presume that this component can be modeled, regardless of its precise structure, like a state machine for purposes of integration with our algorithm: A planning algorithm, P, acts after completion of subtasks, and takes as input the current robot state s_t. It produces as output the index of the next sub-task for execution: $P(s_t) = k$. We then identify the subset of the subtask space $\{T_{i1}, T_{i2}, ...\}$ which can be replaced by operator-driven modules, denoted $\{H_{i1}, H_{i2}, ...\}$ or by autonomous modules denoted by $\{M_{i1}, M_{i2}, M_{i3}\}$ These modules represent parallel task methods implemented as user interface prompts to supplant the autonomous action of the corresponding subsystem.

Definition 1: *The system level of autonomy, A_L is a user-set integer, which determines the number of human-driven modules H_{ij}, that replaces autonomous modules M_{ij} during the execution of a subtask T_i.*

Rather than define A_L as a bulk parameter which accounts for all combinations of manual sub-task activations, we rank the H_{ij} hierarchically as described below, and consider A_L to represent the number of modules rendered active or inactive in this hierarchy.

Establishment of this hierarchy is made by examining the expected failure rates of the autonomous modules, sorting H_{ij} by the probability of failure of its corresponding M_{ij} module. In defining the scale this way, the users choice of A_L is re-cast as a judgement on the relative complexity of the task at hand.

Definition 2: *For each H_{ij}, M_{ij} pair we define an autonomy rank r_ij, as the index in the list of all such pairs when sorted by failure rate of M. All T_k for which no manual module is specified then have r_k = 0.*

Building on the earlier description of the autonomous planning algorithm $P(s_t)$ and letting T_{t+dt} represents the next module the robot will execute, we can propose an augmented process to the state machine selection, and denote it as the Variable Autonomy Planner (VAP).

The operation of the VAP algorithm proceeds as follows: Step 1: The failure rates for each autonomous module are determined experimentally.

Step 2: The appropriate (H_{ij}, M_{ij}) pairs are composed and sorted by the respective failure rate. The order in this list determines r_{ij} for each pair.

Step 3: During execution of the task, the primary planner, P, takes state information from the robot and determines which task to perform next, indicated by the index k.

Step 4: Prior to beginning any task, the operator my, by some interface, change the value of the A_L parameter, in accordance with their judgement of the problem complexity.

Step 5: The level of autonomy is then checked by the augmented planner. If the rank of the current module pair is lower than A_L, then the human-interfaced module is used, and if the rank is greater, then the autonomous module is used.

The VAP algorithm therefor alters the operation of the pre-existing planning algorithm, and it's simplest implementation is summarized below in pseudocode as Algorithm 1.

P$_{aug}$(**s**$_t$,A_L)**:**
  $k \leftarrow$P(s$_t$)
  $r_i \leftarrow$r$_k$
  if $r_i < A_L$ **then**
    $T_{t+dt} \leftarrow H_k$
  if $r_i \geq A_L$ **then**
    $T_{t+dt} \leftarrow M_k$
  return T_{t+dt}

Algorithm 1: Variable Autonomy Planner

III. Methods & Materials

A. Implementation Case

To validate the Augmented Planner, we implement a pick and place operation for item collection and handover. The goal of this task is to collect a specified item from its resting place on a table within the robot's functional workspace using a robot manipulator arm with a two-finger gripper. Items are laid on the table in random 2D poses. Perception is provided as the visual field available to a camera imaging the tabletop workspace. Grasping is achieved by a vertical orientation of the gripper, for which the yaw about the vertical axis allows for finger alignment. We decompose this task into the following sub-task domains: Position Determination, Orientation Estimation, Manipulator Placement, Object Identification, and Object collection.

Within this group of processes, we define six levels at which we can implement variable autonomy, based on supplying human input replacing individual autonomous modules.

1. Full Autonomy: Wherein all actions are performed without human input

2. User sets XY Position: In which the operator specifies the 2D planar coordinates of the item (H_1)

3. User sets Yaw Control: In which the operator specifies the angular orientation of the gripper (H_2)

4. User sets Object Identification: In which the user identifies the object within the visual field (H_3)

5. User sets Object ID & Yaw Control: In which the user specifies the object and sets the grasp angle (H_4)

6. User sets XY Position & Yaw Control: In which the user provides exact coordinates and grasp angle (H_5)

For testing, we investigate three different scenarios: single object detection and handover; multi-object handover; and cluttered environment handover. All cases are tested under both fixed-mode and variable autonomy. We define two performance metrics: completion time and failure rate.

B. Experiment Design

Our trials proceed as follows: twelve subjects are tasked with completing each of the three different problem domains discussed above: single-item, multi-item, and cluttered pick-and-place. Each subject completes six trials in the fixed mode cases, as well as two trials using the VAP setup, for a total of four trials per A_L in the fixed mode for each problem complexity class, and eight trials per complexity level in the VAP-selected case.

In fixed mode trials, the most-failing autonomous modules are replaced with manual operation segments driven through the UI; in the sliding-scale tests, the user is allowed to dynamically adjust the autonomic components between trials. Four total items are presented to the robot, as shown in Figures 3 & 4, and these items are randomly placed and aligned. Additional objects are used for the cluttered test, but not manipulated.

For the single-item trials the user is directed to use the interface to grasp the object with the robot for handover. In the multi-item case, a random pool of three items is selected, and placed in the workspace (shown in Figure 1, right). The operator is directed to pick one item for handover. This task is then repeated for one of the remaining items. In the final task, cluttered space picking, all items are placed along with additional non-task items. The user is directed to collect two of the items from the workspace for handover, as in the prior test.

In each trial of the above experiments, we record both the number of times the task fails and the time to completion, representing accuracy and speed respectively. Our analysis

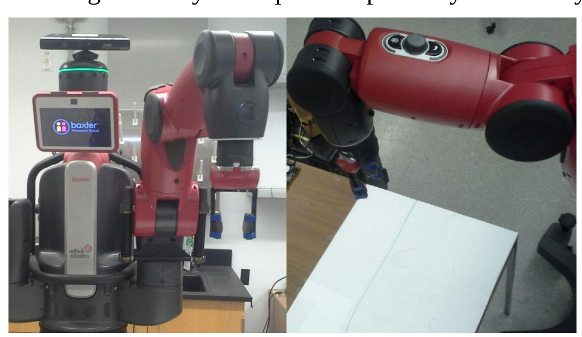

Fig. 1. Baxter: Imaging Pose (Left) and Workspace (Right)

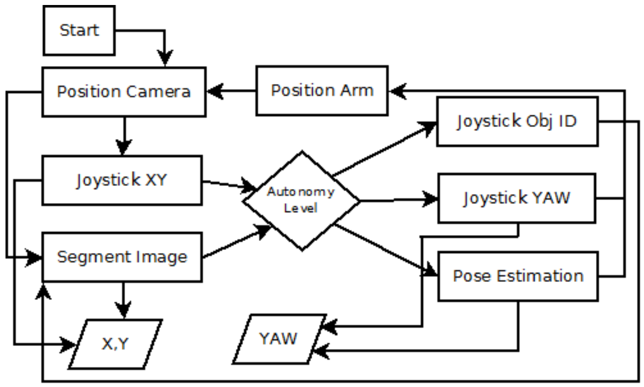

Fig. 2. Main Software Loop

hinges on examination of the impact of the A_L parameter on these two metrics in the fixed-mode trials to support our assertions regarding the relative merits of the robot and human's capabilities, and on the comparison of average performance characteristics of the variable autonomy case versus the fixed-mode case.

Our hypothesis includes the statement that the autonomy level at which *optimal performance* is achieved may be context dependent. For systemic investigations in which multiple parameters are measured, it can be difficult to assign weighting values to each parameter to find a balance between contrasting objectives. For this experiment, we elect to use the following joint metric: for any complexity level battery, we claim that the A_L for which the error rate is at its minimum level across all autonomy levels, and at which the average completion time is minimized for those lowest failure rate levels, constitutes the 'optimal' A_L.

C. Robot Hardware

For these experiments, we implement pick-and-place functionality on a ReThink Robotics Baxter Research Robot (Figure 1, left). With regards to safe operation, the Baxter unit employs a fully back-drivable set of actuators in its manipulator arm, meaning that the robot can be easily manipulated by external force, such as if an operator needs to resist its motion in a collision.

Baxter is also equipped with a wrist-mounted camera. We used this camera for the recognition and detection of objects in the workspace. In addition, we provide a raw video stream (Figure 3, right) from this sensor to the user's teleoperation workstation during manual positioning and grasping contexts, to augment the operator's perspective, which may be limited by distance from the robot's workspace.

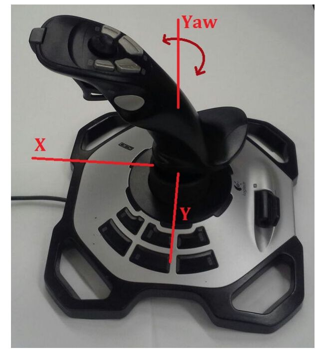

Fig. 4. Joystick deployed for user interface, illustrating cartesian control and yaw rotation control axis orientations

A block-diagram of the primary process which implements module-exchange between the autonomous components and the manual components is shown in Figure 2. The level of autonomy acts as an additional state variable for P. In our software, the VAP is thus introduced precisely as a new variable to the pre-existing state machine, for which the functional block is contains Algorithm 1.

D. Perception Software

For the autonomous modules, we have constructed a fully independent pick and place system, comprised of the Baxter's ROS based inverse kinematics and joint position control modules, and machine vision perception components leveraging OpenCV bindings for python.

To implement object detection, a background filtering method was used, with segmentation based on marching squares blob-detection. Pose estimation is performed by comparison of bounding box area to object area. Tracking estimates are made by calculating the visual center of mass of the object, and projecting pixel coordinates onto the workspace. Figure 3 shows diagnostic output of these tracking systems (left image).

E. User Interface

There are three modes in which user input is taken from the operator: Object Identification, XY Positioning, and Yaw orientation. Each of these is driven by input from a USB Joystick, interfaced to supply X and Y motions, Yaw rotation about its main axis, and button input from its trigger. The joystick used is shown in Figure 4, with labels for the local X, Y, Yaw control coordinate frame.

The XY positioning and Object Identification components are identical in appearance to the user, except for prompting.

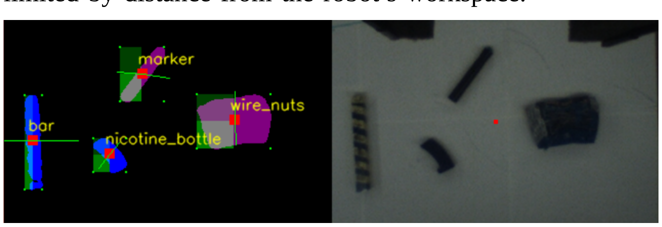

Fig. 3. Object recognition, tracking and pose estimation display; diagnostic presentation for autonomous modes (left), video feed for operator (right)

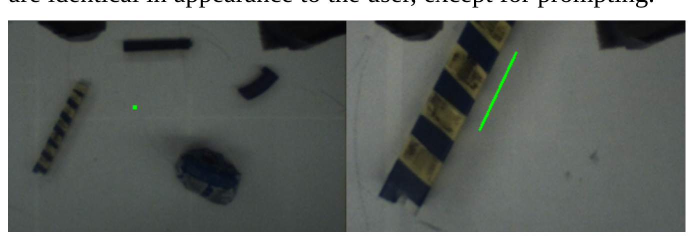

Fig. 5. User interface: XY/Object ID GUI (Left), Yaw selection GUI (Right)

Both instances provide a real-time feed of video from the wrist cameras, and allow the user to position a target indicator in the field of view as shown in Figure 5 (Left). In the XY positioning case, the registered point is converted to workspace coordinates, and in the Object Identification mode, the nearest located item found in the visual field is tracked item for retrieval.

The second mode, as seen in Figure 5 (Right), is the Yaw-set mode. In this module, the user is able to set the orientation of the manipulator. A line is projected onto the video feed, and when the user twists the joystick about the vertical axis the line rotates about a fixed point at the top of the image. The projected orientation of the line matches the orientation of the gripper after rotation.

IV. Results

To begin with, we can examine the relationship within the fixed-mode data as shown in Tables I & II, which describe the average task completion times and failure rates as a function of the A_L parameter. These data are also represented graphically in Figure 6.

The first observation we can make is that there is a significant difference in task completion time between fully-automatic execution and execution with manual components. The average completion time over all autonomous cases (Table I, column A_L: 1), is 32 seconds, whereas the completion time average over $A_L > 1$ is 57 seconds; a 78% decrease in speed per item.

However, comparing trials involving any degree of manual intervention, we find that the range of completion times do not vary significantly, with the average deviation over all fixed mode settings and complexity classes being 5.4 seconds (10.3% of the bulk mean), and the largest recorded deviation being 10.9 seconds.

This abrupt change in speed between autonomous and intervention modes, without significant deviation among intervention modes, offers support for our proposition that human interventions tend to be more time-intensive.

In addition, there is also a small advantage in accuracy. For both the single- and mixed-item cases, the failure rate

Setting	A_L: 1	2	3	4	5	6
Single	32 (0.6)	58 (2.2)	56 (6.9)	55 (9.5)	56 (6.5)	61 (7.6)
Multi	34 (1.5)	64 (10.9)	49 (5.6)	59 (7.6)	65 (2.6)	58 (7.1)
Cluttered	29 (1.7)	51 (3.1)	56 (3.2)	55 (2.3)	52 (7.9)	61 (6.9)

TABLE I
Completion Time for Fixed-mode Trials: average(variance)

Setting	A_L :	1	2	3	4	5	6
Single		10%	5%	0%	5%	0%	0%
Multi		10%	10%	0%	5%	0%	0%
Cluttered		45%	20%	10%	15%	5%	5%

TABLE II
Failure Rates (Fixed-Mode Trials)

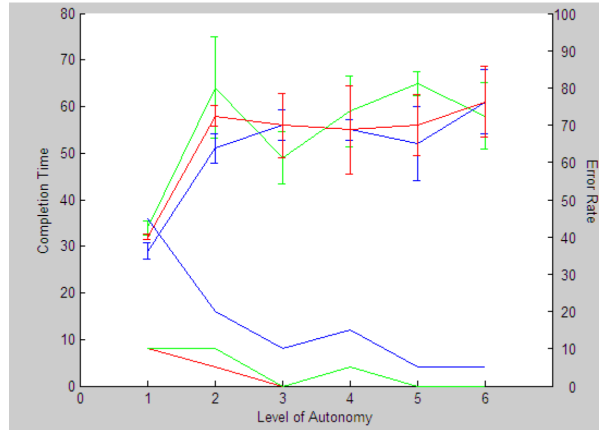

Fig. 6. Comparison of failure rate and average completion time versus level of autonomy for all task domains (single-item in red, multi-item in green, cluttered in blue) error bars representing sample variance are included for the completion time data; error rates are cumulative over all trials, and thus include no variance

for $A_L = 1$ is 10% (shown in Table II). If we average over the same cases for the manual modes, we find failure rates of 2% and 3%, respectively. In the cluttered case, this pattern is even more pronounced- the fully autonomous mode's failure rate 45%, and the fault rate curve over the level of autonomy scale decreases as autonomy increases, down to 5% for fully manual operation.

This relationship is illustrated more succinctly in Figure 6, which plots both failure rates (right hand axis) and completion times (left hand axis) on a common graph. These experiments illustrate that the conditions of our hypothesis hold- that autonomy and intervention are complimentarily related in these two metrics.

Another important piece of confirming evidence is towards our proposition that we have proposed tasks exhibiting differing levels of complexity. The failure rates for the single- and mixed-item cases are very similar, but those for the cluttered case are uniformly higher. This verifies that we are indeed changing the context of the test cases over these problem classes, in the sense of presenting an increasingly challenging objective to the human/robot system.

Our results also confirm that the best performance is achieved through a shared-control model: in the single-item case, we find that $A_L = 3$ is the level at which the failure rate is lowest (0%) and the speed is highest, and likewise for the multi-item case. For the cluttered case, however, the highest-speed/least-error performance is achieved at $A_L = 5$. In none of these trials was the optimum performance achieved in either the fully autonomous or the fully manual mode.

As the complexity of the task increases, we see a shift in

Setting	Faults	Time per item	Favored A_L
Single	0%	50s	3
Multi	0%	46s	3
Cluttered	5%	51s	5

TABLE III
Variable Autonomy Results

the level of autonomy associated with the optimal performance. This indicates that the ideal A_L is indeed context-dependent over the chosen experiment domains.

When we look to the performance of the Variable Autonomy Planner (indicated in Table III), we find that the cumulative error rate is reduced to 0% in the single- and multi-item cases, and 5% in the cluttered item cases.

Additionally, across all three domains, the variable autonomy method exhibits marginally faster completion time than in the fixed-mode case, though still less quickly than the autonomous case. On average, this difference amounts to a 9% speed increase over the corresponding fixed mode case, and a 5% decrease relative to the autonomous speed. This is outside the variance for the single- and multi-item cases, but within it for the cluttered case. Determination of whether this discrepancy is an artifact or a true pattern is a topic of ongoing experimentation.

In the single- and multi-item cases, users primarily selected to the Yaw setting ($A_L = 3$), and in the cluttered experiment users tended to select Object Identification and Yaw control ($A_L = 5$), leaving only XY positioning to the perception system.

In the fixed-mode cases, these A_L are the same as were determined to be the sought-after optima, indicating that the users were able to select the optimal autonomy level. Because the user-selected level both changes with problem conditions and matches the fixed-case optima, this constitutes evidence that the user constitutes a good judge for selecting the A_L parameter dynamically in response to changing task conditions.

V. Conclusions & Future Work

In this paper we discussed the impact of human and robot collaborative interaction on the efficiency and efficacy of task completion, with a specific emphasis on the relation between the relative advantages between the human and machine.

Further, we postulated that the effectiveness of a shared control system may be context-dependent, and as such, a system which adapts the level of autonomy would out perform a system with a fixed level of autonomy. We also proposed that this adaptation task be assigned to the human as a sliding scale input, resulting in the Variable Autonomy Planner for augmenting an autonomous system.

To investigate these hypotheses we posed an example pick-and-place task and devised a set of scenarios designed to imitate such variable conditions. We tested this system in the typical fixed mode, as well as the adjustable mode, and from these experiments observed that our framing assumptions held for this problem case, and our hypothesis about the relative relationships between quality and speed versus level of autonomy was supported.

We determined that the Variable Autonomy Planner was effective, with the users correctly identifying the optimal levels of autonomy determined in the fixed-mode experiments.

Additional experiments investigating non-environmental aspects operation, such as human factors, would be an important step towards generalizing these results. On particular case is the investigation of the apparent small advantage in speed for the VAP execution batteries.

We are also beginning to investigate context-dependent changed in non-complexity oriented changes in task domain, such as single-module failure rate increasing factors. One example of this would be changes in light conditions, which almost exclusively impacts the object recognition module.

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References

[1] Doelling, K., Shin, J., and Popa, D. O., *Service robotics for the home: a state of the art review,*", Proceedings of the 7th International Conference on PErvasive Technologies Related to Assistive Environments , 35, ACM 2014.

[2] Christina Rodel, Susanne Stadler, Alexander Meschtscherjakov, Manfred Tscheligi; *Towards Autonomous Cars: The Effect of Autonomy Levels on Acceptance and User Experience*, Proceedings of the 6th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (pp. 1-8), ACM, 2014.

[3] Clint Heyer *Human-Robot Interaction and Future Industrial Robotics Applications*, Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on. IEEE, 2010.

[4] Sumit Kumar Das, Ankita Sahu and Dan O. Popa *Mobile App for Human-Interaction with Sitter Robots*, Smart Biomedical and Physiological Sensor Technology XIV (Vol. 10216, p. 102160D). International Society for Optics and Photonics, 2017.

[5] Fong, T.; Thorpe, C.; Baur, C. *Collaborative Control: A robot-centered model for vehicle teleoperation* Proceedings of the AAAI Spring Symposium on Agents with Adjustable Autonomy. Menlo Park, California: AAAI, 1999

[6] Bernard P. Zeigler; *High Autonomy Systems: Concepts and Models*, AI, Simulation and Planning in High Autonomy Systems, Proceedings. (pp. 2-7), IEEE, 1990.

[7] Krishnanand N. Kaipa, Akshaya S. Kankanhalli-Nagendra, Nithyananda B. Kumbla; *Enhancing Robotic Unstructured Bin-Picking Performance by Enabling Remote Human Interventions in Challenging Perception Scenarios*, 2016 IEEE International Conference on Automation Science and Engineering (CASE), IEEE, 2016.

[8] Krishnanand N. Kaipa, Srudeep Somnaath Thevendria-Karthic, Shaurya Shriyam, Ariyan M. Kabir, Joshua D. Langsfeld, and Satyandra K. Gupta; *Resolving Automated Perception System Failures in Bin-Picking Tasks Using Assistance from Remote Human Operators*, 2015 IEEE International Conference on Automation Science and Engineering (CASE), IEEE, 2015.

[9] Michael A. Goodrich, Dan R. Olsen Jr., Jacob W. Crandall and Thomas J. Palmer, *Experiments in Adjustable Autonomy*, Proceedings of IJCAI Workshop on Autonomy, Delegation and Control: Interacting with Intelligent Agents. Seattle, WA: American Association for Artificial Intelligence Press, 2001.

[10] Wolf, M. T., Assad, C., Vernacchia, M. T., Fromm, J., Jethani, H. L. *Gesture-based robot control with variable autonomy from the JPL BioSleeve*, IEEE International Conference on Robotics and Automation (ICRA), (pp. 1160-1165); 2013.

[11] Beer, J. M., Fisk, A. D., Rogers, W. A. *Toward a framework for levels of robot autonomy in human-robot interaction*, Journal of Human-Robot Interaction, 3(2), 74; 2014.

[12] Fitts, Paul M. *The information capacity of the human motor system in controlling the amplitude of movement*, Journal of Experimental Psychology. 47 (6), (pp.381391), 1954